Help Validate Chronoacupuncture for Jet Lag

Introduction

Jet lag is a problem for many international travelers. Traditional interventions like sunrise viewing, strategic melatonin use, sleep strategy and meal scheduling show variable efficacy.[1] But acupuncturists propose a different approach.

Nearly 50 years ago, Khoe[2] and then Amaro in 2001[3] postulated that stimulating horary points based on destination time zones during travel could help pre-synchronize organ systems to the destination time zone, targeting jet lag's circadian misalignment. Theory and anecdotal evidence say this works. Unfortunately, no studies have validated it. We aim to address that lack of evidence with an app that provides a destination-based 24-hour schedule of acupuncture points for travelers to stimulate as they travel.

We invite readers planning international travel to participate.

Circadian Biology and the Western Understanding of our Internal Clocks

The scientific literature describes how circadian rhythms synchronize physiological processes to a 24-26 hour cycle through the suprachiasmatic nucleus (SCN), a cluster of neurons entrained by light.[4,5] The SCN coordinates peripheral clocks in organs like the liver and heart and tissue-specific functions such as vascular and blood pressure regulation.[6,7] Then, clock genes drive a molecular feedback loop, producing rhythmic gene expression.[8] This synchronization is maintained by zeitgebers, external signals such as light, temperature, activity, and meals. [9]

Jet lag results when rapid time zone changes desynchronize these internal clocks.

A 2024 comprehensive review shows that acupuncture significantly affects circadian biology, demonstrating measurable impacts on circadian clock genes, neurotransmitter regulation, and sleep-wake cycles.[10] While these studies employed acupuncture points known to assist with sleep, they establish the neurobiological plausibility of acupuncture serving as a non-photic zeitgeber for circadian rhythm modulation.

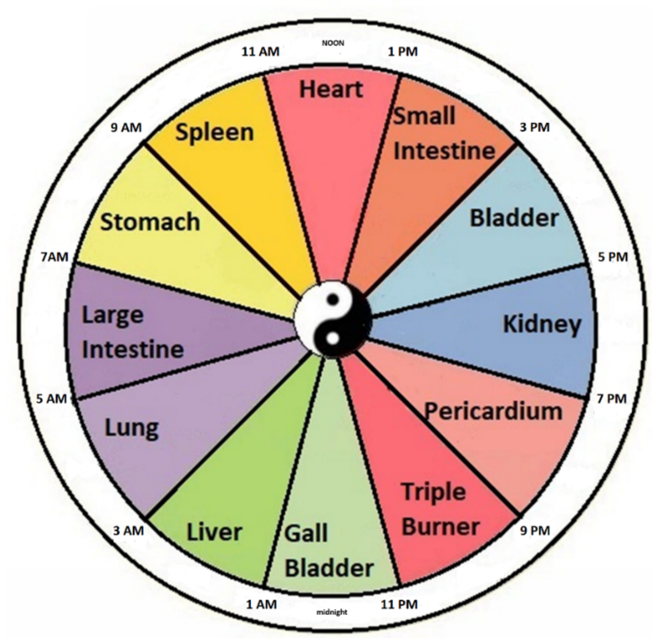

Figure 1: *24-hour Organ Clock*

Table 1: 24-hour Schedule

Time Range	Meridian / Organ System	Horary -> Suggested Point
3:00 AM – 5:00 AM	Lung	LU-8
5:00 AM – 7:00 AM	Large Intestine	LI-1
7:00 AM – 9:00 AM	Stomach	ST-36
9:00 AM – 11:00 AM	Spleen	SP-3 -> SP-10*
11:00 AM – 1:00 PM	Heart	HT-8
1:00 PM – 3:00 PM	Small Intestine	SI-5
3:00 PM – 5:00 PM	Bladder	BL-66 -> BL-2*
5:00 PM – 7:00 PM	Kidney	KI-10 -> KI-27*
7:00 PM – 9:00 PM	Pericardium	PC-8
9:00 PM – 11:00 PM	San Jiao (Triple Warmer)	SJ-6
11:00 PM – 1:00 AM	Gallbladder	GB-41 -> GB-20*
1:00 AM – 3:00 AM	Liver	LIV-1 -> LIV-8*

Table 1
*The * suggested points are modifications implemented for practical accessibility.*

Point Changes for Practical Accessibility

While Amaro recommended stimulating the horary points on each meridian, initial testers of the app who were flying economy class were hesitant to remove shoes to stimulate ankle and foot points. Our solution was to use alternative channel points to replace “inaccessible” points understanding that it would be better to stimulate the channel with a “non-ideal” yet accessible point rather than skip the channel altogether.

For practicality and ease of location, we chose to use SP-10 (above the knee) for SP-3, BL-2 (inner eyebrow) for BL-66, GB-20 (back of head) for GB-41, LIV-8 (medial knee) for LIV-1, and KI-27 (upper chest) for KI-10. In TCM theory, any point on a meridian can affect the entire pathway, and this substitution maintains meridian specificity.

How to participate

On your iPhone, go here https://jetlagpro.com/#download to get the app. Detailed instructions are provided on the website. This research app is entirely free with no add-ons, upgrades or advertisements. As an incentive, participants who use the app for a valid journey and complete the survey will enjoy free lifetime access when the app is officially released.

Once you have the app installed, follow the simple directions to receive timing and instructions. If you have questions, email us info@jetlagpro.com.

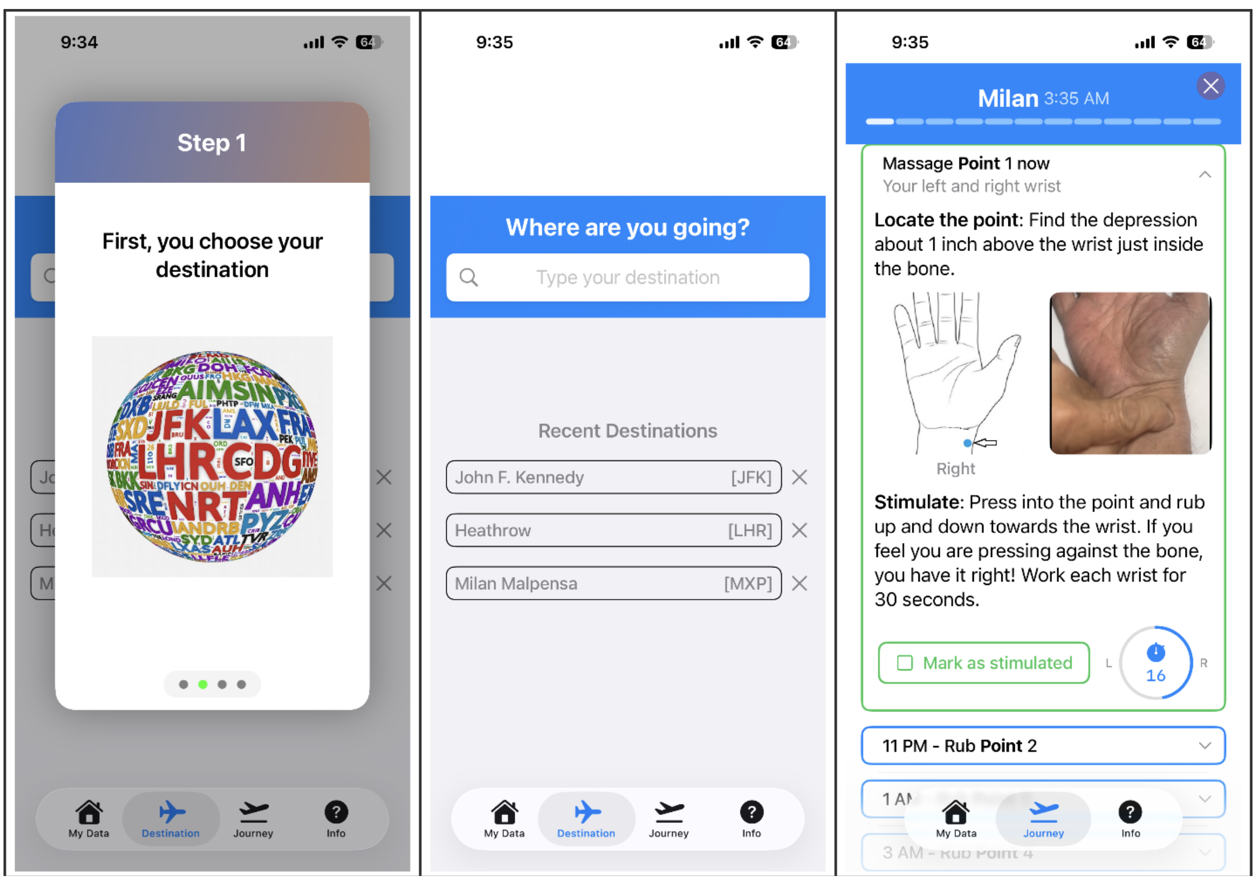

Figure 2 – The Three Steps

Scheduled reminders pop up over the next 24 hours to prompt the traveler to do the next point. The reminders are persistent and remain on the user's phone (or Apple Watch) until clicked.

When clicked, the traveler is shown the correct point image beside a looping video showing how to stimulate the point.

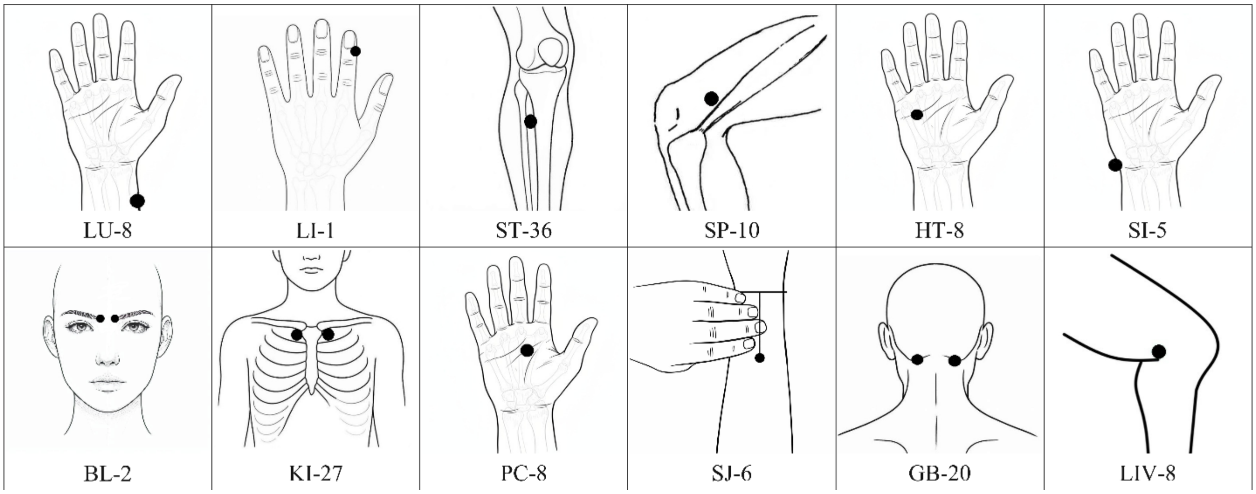

Figure 3 – Point Images

After the trip ends, the app saves point stimulation data along with origin and arrival time zones to a central database. Then a survey reminder is scheduled for 2 days later. This survey is a streamlined assessment based on the Liverpool Jet Lag Questionnaire's core symptom domains.

Data Analysis

Our primary question is "Does chronoacupuncture (as provided by the app) reduce jet lag symptoms as compared to no intervention?" We will compare survey results to baseline severity data from Waterhouse et al.'s 2007 comprehensive quantitative assessment[1]. Participants will be grouped by the number of acupressure points stimulated along with the number of time zones crossed. This study is pre-registered at Open Science Framework: osf.io/jm4w7.

Discussion

Traditional clinical trials face significant challenges in jet lag research because they cannot control countless travel variables, including demographics and environmental factors. In this study, these uncontrolled variables do not undermine validity because the expected sample size of point usage data and survey results will provide sufficient statistical power to detect meaningful effects.

References

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